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BEAMFORMING APPARATUS AND METHOD

Abstract

A beamforming apparatus and method are provided, to implement fully connected beamforming by using a wavelength division multiplexing technology. In this application, radio frequency signals are modulated on laser signals with different wavelengths; and then after combination and splitting, phase adjustment is selectively performed on the radio frequency signals with the different wavelengths on paths through wavelength selection units, to change transmit angles of beams transmitted at antenna ends. In another manner, in this application, radio frequency signals are modulated on laser signals with different wavelengths, each path of optical signal in the modulated radio frequency signal is split into a plurality of paths for phase adjustment, and then the phase-adjusted signals are combined and sent, to change transmit angles of beams transmitted at antenna ends. In addition, the solutions provided in this application are applicable to direct modulation, direct detection, and heterodyne or homodyne coherent detection.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of International Application No. PCT/CN2022/133533, filed on Nov. 22, 2022, the disclosure of which is hereby incorporated by reference in its entirety.

TECHNICAL FIELD

[0002] This application relates to the field of optical communication technologies, and in particular, to a beamforming apparatus and method.

BACKGROUND

[0003] In a wireless communication network, a beamforming technology has become a very key technology, and is used to improve a signal-to-noise ratio, a transmission distance, and the like of a link signal. A current mainstream technical solution is to use an electrical device to perform phase control, to implement a beamforming function. The solution is, for example, a digital phase shifting solution, an analog phase shifting solution, or a hybrid phase shifting solution of a baseband chip.

[0004] With development of wireless communication technologies, a wireless frequency band is gradually shifting toward a higher frequency band, and a requirement for signal capacity is increased, leading to an increase in a quantity of antennas. Currently, a subarray-based design solution is used to meet a requirement for increased antenna data. However, in the subarray-based design solution, power consumption is high, and a quantity of antennas allocated to each subarray is limited. Consequently, it is difficult to obtain a high-quality beam.

SUMMARY

[0005] Embodiments of this application provide a beamforming apparatus and method, to implement fully connected beamforming by using a wavelength division multiplexing technology.

[0006] According to a first aspect, an embodiment of this application provides a beamforming apparatus. The apparatus includes a first laser signal source, a plurality of modulators, a first optical splitting device, a plurality of wavelength selection units, and a plurality of phase shifters. The laser signal source is configured to transmit a plurality of laser signals with different wavelengths. The plurality of modulators respectively modulate to-be-sent radio frequency signals on the laser signals with the corresponding wavelengths to obtain first modulated optical signals. The first optical splitting device combines the plurality of first modulated optical signals, and then distributes the combined first modulated optical signals to a plurality of optical paths. Each optical path corresponds to a second modulated optical signal. Each optical path corresponds to one wavelength selection unit. The wavelength selection unit receives, on the corresponding optical path, the second modulated optical signal from the first optical splitting device, and selectively transmits an optical signal with a corresponding wavelength in the received second modulated optical signal to a phase shifter. Each of the plurality of phase shifters performs phase shifting (that is, phase adjustment) on a received optical signal with a corresponding wavelength, and then transmits the optical signal to a corresponding connected wavelength selection unit. The wavelength selection unit receives optical signals with a plurality of wavelengths from a plurality of phase shifters, and fuses the optical signals to obtain a third modulated optical signal for transmission.

[0007] In this embodiment of this application, a full-connection method of light is used, to implement effective routing of light through wavelength multiplexing and mode multiplexing, so as

to implement expansion of a traffic channel. In addition, a quantity of used modulators is only related to a quantity of full connections, and complexity is low, which can reduce a size of the beamforming apparatus to some extent. In addition, lasers with different wavelengths are used to correspondingly transmit beam signals at different antenna ends, so that frequencies of radio frequency signals do not need to be adjusted; and phase shifting is performed on a signal with a corresponding wavelength on each path through a phase shifter, so that a transmit angle of the radio frequency signal that is transmitted at a radio frequency antenna end and that is modulated at the wavelength can be changed. In this way, beamforming of each radio frequency signal is implemented.

[0008] In some possible implementations, the beamforming apparatus further includes a plurality of detectors and a plurality of radio frequency antenna units. Each of the plurality of detectors corresponds to one wavelength selection unit, and is configured to: receive a third modulated optical signal from the corresponding wavelength selection unit, perform optical-to-electrical conversion on the received third modulated optical signal to obtain an electrical signal, and send the electrical signal through a radio frequency antenna unit corresponding to the first detector.

[0009] An example in which the beamforming apparatus includes M modulators, N wavelength selection units, and $N*M$ phase shifters is used.

[0010] The first laser signal source is configured to transmit M laser signals, where wavelengths of the M laser signals are different. The M modulators are in a one-to-one correspondence with the M laser signals, and are respectively configured to modulate radio frequency signals on the corresponding laser signals to obtain M first modulated optical signals, where radio frequency signals modulated on any two of the M laser signals are the same or different. The first optical splitting device is coupled to the M modulators; and is configured to combine and distribute the M first modulated optical signals into N optical paths to obtain N second modulated optical signals corresponding to the N optical paths. The N wavelength selection units are coupled to the first optical splitting device. The N wavelength selection units are in a one-to-one correspondence with the N second modulated optical signals. A first wavelength selection unit is configured to: receive the second modulated optical signal sent by the first optical splitting device, and send signals with M wavelengths in the second modulated optical signal to M phase shifters in a one-to-one correspondence; and the first wavelength selection unit is any one of the N wavelength selection units. The M phase shifters are respectively configured to: adjust phases of the received optical signals, and send the phase-adjusted optical signals to the first wavelength selection unit. The N wavelength selection units are coupled to N detectors in a one-to-one correspondence. The first wavelength selection unit is further configured to: fuse the received optical signals from the corresponding M phase shifters to obtain a third modulated optical signal, and transmit the third modulated optical signal.

[0011] In some possible implementations, the beamforming apparatus further includes the N detectors and N radio frequency antenna units. The N detectors are coupled to the N wavelength selection units in a one-to-one correspondence, and the N detectors are coupled to the N radio frequency antenna units in a one-to-one correspondence. The N detectors are configured to: receive third modulated optical signals sent by the corresponding wavelength selection units, perform optical-to-electrical conversion on the received third modulated optical signals to obtain electrical signals, and send the electrical signals through the radio frequency antenna units corresponding to the first detectors.

[0012] In a possible design, the wavelength selection unit is a reconfigurable optical add-drop multiplexer ROADM.

[0013] In a possible design, the beamforming apparatus further includes: a control unit, configured to control phase shift spacings of the $N*M$ phase shifters.

[0014] The control unit controls each phase shifter to adjust a phase shift variation amount of an optical signal with a corresponding wavelength, in other words, may change a transmit angle of the

radio frequency signal with the wavelength at an antenna end after beamforming. In this embodiment of this application, a quantity M of radio frequency signals that can be simultaneously transmitted is not specifically limited. M radio frequency signals may be transmitted in a time division multiplexing manner, or may be all simultaneously transmitted, to implement full connection of the M radio frequency signals.

[0015] In a possible design, the first laser signal source includes M first laser devices, and the M first laser devices are coupled to the M modulators in a one-to-one correspondence. Wavelengths of the M first laser devices are different.

[0016] In a possible design, the first laser signal source includes an M -wavelength laser device and a wavelength division multiplexer, the M -wavelength laser device is coupled to the wavelength division multiplexer, and the wavelength division multiplexer is coupled to the M modulators.

[0017] In a possible design, the beamforming apparatus further includes a second laser signal source and a second optical splitting device.

[0018] The second laser signal source is configured to transmit M carrier optical signals that are in a one-to-one correspondence with the M laser signals.

[0019] The second optical splitting device is configured to couple and distribute the M carrier optical signals to the N detectors.

[0020] Each of the N detectors is specifically configured to: perform coherent detection on the received third modulated optical signal based on received carrier optical signals, and perform optical-to-electrical conversion on the coherently detected third modulated optical signal to obtain an electrical signal.

[0021] In the foregoing design, heterodyne coherent detection is implemented by adding the second laser signal source and the second optical splitting device, so that the design may be applied to an application scenario of heterodyne coherent detection. A radio frequency of each of M beams sent by radio frequency antenna ends is related to a frequency of a laser signal with each wavelength in the first laser signal source and a frequency of a laser signal with each wavelength in the second laser signal source. In some scenarios, a frequency of a laser signal with each wavelength in the first laser signal source or the second laser signal source may be further changed, to change a radio frequency of the laser signal, with the corresponding wavelength, sent by the radio frequency antenna end, to implement a frequency conversion operation.

[0022] In a possible design, the beamforming apparatus further includes a third optical splitting device.

[0023] One end of the third optical splitting device is coupled to the first laser signal source, the other end of the third optical splitting device is coupled to the N detectors, and the third optical splitting device is configured to couple and distribute the M laser signals to the N detectors. A first detector in the N detectors is specifically configured to: perform coherent detection on a received fourth modulated optical signal based on received laser signals, and perform optical-to-electrical conversion on a coherently detected fourth modulated optical signal to obtain an electrical signal.

[0024] In the foregoing design, homodyne coherent detection may be implemented by adding the third optical splitting device, so that the design may be applied to an application scenario of homodyne coherent detection. A radio frequency of each of M beams sent by radio frequency antenna ends is related to a frequency of a laser signal with each wavelength in the first laser signal source.

[0025] According to a second aspect, an embodiment of this application provides another beamforming apparatus. The apparatus includes a first laser signal source, M modulators, M first beam splitters, N first wavelength division multiplexers, and $N \times M$ phase shifters. The first laser signal source is configured to transmit M laser signals, where wavelengths of the M laser signals are different.

[0026] The M modulators are in a one-to-one correspondence with the M laser signals, and are respectively configured to modulate radio frequency signals on the corresponding laser signals to

obtain M first modulated optical signals, where radio frequency signals modulated on any two of the M laser signals are the same or different.

[0027] The M first beam splitters are coupled to the M modulators in a one-to-one correspondence; and each of the M first beam splitters is coupled to input ends of N phase shifters, and is configured to: receive a first modulated optical signal sent by a corresponding modulator, distribute the received first modulated optical signal into N second modulated optical signals, and distribute the N second modulated optical signals to the N phase shifters in a one-to-one correspondence.

[0028] The N phase shifters are respectively configured to adjust phases of the received second modulated optical signals, to obtain third modulated optical signals.

[0029] An input end of each of the N first wavelength division multiplexers is coupled to output ends of M phase shifters, different first beam splitters are respectively coupled to input ends of M phase shifters coupled to a same first wavelength division multiplexer, and each first wavelength division multiplexer is configured to couple third modulated optical signals from the M phase shifters into a fourth modulated optical signal.

[0030] In a possible implementation, the beamforming apparatus further includes N detectors and N radio frequency antenna units. Second ends of the N first wavelength division multiplexers are coupled to input ends of the N detectors in a one-to-one correspondence, and output ends of the N detectors are coupled to the N radio frequency antenna units in a one-to-one correspondence. Each of the N detectors is configured to: receive a fourth modulated optical signal from a corresponding first wavelength division multiplexer, perform optical-to-electrical conversion on the fourth modulated optical signal to obtain an electrical signal, and send the electrical signal through a corresponding radio frequency antenna unit.

[0031] Compared with the beamforming apparatus provided in the first aspect, the beamforming apparatus provided in the second aspect does not require participation of a wavelength selection unit, and therefore costs and a size of the beamforming apparatus can be reduced.

[0032] In a possible design, the beamforming apparatus further includes: a control unit, configured to control phase shift spacings of the $N \times M$ phase shifters.

[0033] In a possible design, the first laser signal source includes M first laser devices, and the M first laser devices are coupled to the M modulators in a one-to-one correspondence. Wavelengths of the M first laser devices are different.

[0034] In a possible design, the first laser signal source includes a M-wavelength laser device and a second wavelength division multiplexer, the M-wavelength laser device is coupled to the second wavelength division multiplexer, and the second wavelength division multiplexer is coupled to the M modulators.

[0035] In a possible design, the beamforming apparatus further includes a second laser signal source and a second optical splitting device. The second laser signal source is configured to transmit M carrier optical signals that are in a one-to-one correspondence with the M laser signals.

[0036] The second optical splitting device is configured to couple and distribute the M carrier optical signals to the N detectors.

[0037] Each of the N detectors is specifically configured to: perform coherent detection on the received fourth modulated optical signal based on received carrier optical signals, and perform optical-to-electrical conversion on the coherently detected fourth modulated optical signal to obtain an electrical signal.

[0038] In a possible design, the beamforming apparatus further includes a third optical splitting device.

[0039] One end of the third optical splitting device is coupled to the first laser signal source, the other end of the third optical splitting device is coupled to the N detectors, and the third optical splitting device is configured to couple and distribute the M laser signals to the N detectors.

[0040] A first detector in the N detectors is specifically configured to: perform coherent detection on a received fourth modulated optical signal based on received laser signals, and perform optical-

to-electrical conversion on a coherently detected fourth modulated optical signal to obtain an electrical signal.

[0041] According to a third aspect, an embodiment of this application provides a beamforming method. The method includes: transmitting M laser signals with different wavelengths; respectively modulating radio frequency signals on the M laser signals to obtain M first modulated optical signals, where radio frequency signals modulated on any two of the M laser signals are the same or different; combining and distributing the M first modulated optical signals into N optical paths, to obtain N second modulated optical signals corresponding to the N optical paths; and on each optical path, performing phase adjustment on optical signals with M wavelengths included in the second modulated optical signal, coupling and performing optical-to-electrical conversion on the M phase-shifted optical signals to obtain an electrical signal, and transmitting the electrical signal through an antenna.

[0042] In a possible design, the method further includes: transmitting M carrier optical signals that are in a one-to-one correspondence with the M laser signals; and combining and splitting the M carrier optical signals into N paths of optical signals, where each of the N paths of optical signals includes M carrier optical signals. The coupling and performing optical-to-electrical conversion on the M phase-shifted optical signals on each path to obtain an electrical signal includes: on each optical path, respectively performing coherent detection on the M phase-shifted optical signals based on M received carrier optical signals, and performing optical-to-electrical conversion on the coherently detected optical signals to obtain the electrical signal.

[0043] In a possible design, the method further includes: coupling and splitting the M laser signals with the different wavelengths into N paths of optical signals, where each of the N paths of optical signals includes M laser signals with the different wavelengths. The coupling and performing optical-to-electrical conversion on the M phase-shifted optical signals to obtain an electrical signal includes: respectively performing coherent detection on the M phase-shifted optical signals based on M received laser signals, and performing optical-to-electrical conversion on the coherently detected optical signals to obtain the electrical signal.

[0044] According to a fourth aspect, an embodiment of this application provides a beamforming method. The method includes: transmitting M laser signals with different wavelengths; respectively modulating radio frequency signals on the M laser signals to obtain M first modulated optical signals, where radio frequency signals modulated on any two of the M laser signals are the same or different; splitting each of the M first modulated optical signals into N second modulated optical signals, to obtain $N \times M$ second modulated optical signals; respectively performing phase adjustment on each of the $N \times M$ second modulated optical signals to obtain a third modulated optical signal, to obtain $N \times M$ third modulated optical signals; coupling M third modulated optical signals included in each of N third modulated optical signal groups in the $N \times M$ third modulated optical signals into a fourth modulated optical signal, to obtain N fourth modulated optical signals, where wavelengths of the M third modulated optical signals included in each of the N third modulated optical signal groups are different; and performing optical-to-electrical conversion on each of the N fourth modulated optical signals to obtain an electrical signal, and transmit the electrical signal through a corresponding antenna.

[0045] In a possible design, the method further includes: transmitting M carrier optical signals that are in a one-to-one correspondence with the M laser signals; and combining and splitting the M carrier optical signals into N paths of optical signals, where each of the N paths of optical signals includes M carrier optical signals. The performing optical-to-electrical conversion on each of the N fourth modulated optical signals to obtain an electrical signal includes: performing coherent detection on the received fourth modulated optical signal based on M received carrier optical signals, and performing optical-to-electrical conversion on the coherently detected fourth modulated optical signal to obtain the electrical signal.

[0046] In a possible design, the method further includes: coupling and splitting the M laser signals

with the different wavelengths into N paths of optical signals, where each of the N paths of optical signals includes M laser signals with the different wavelengths, and the N paths of optical signals are in a one-to-one correspondence with the N fourth modulated optical signals. The performing optical-to-electrical conversion on each of the N fourth modulated optical signals to obtain an electrical signal includes: performing, based on M laser signals in a first path of optical signals, coherent detection on a fourth modulated optical signal corresponding to the first path of optical signals, and performing optical-to-electrical conversion on the coherently detected optical signal to obtain an electrical signal.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0047] FIG. 1 is a diagram of a structure of an integrated optical waveguide beamforming apparatus;

[0048] FIG. 2A is a diagram of a structure of a beamforming apparatus according to an embodiment of this application;

[0049] FIG. 2B is a diagram of a structure of another beamforming apparatus according to an embodiment of this application;

[0050] FIG. 3 is a schematic flowchart of beamforming based on a beamforming apparatus according to an embodiment of this application;

[0051] FIG. 4 is a diagram of a structure of another beamforming apparatus according to an embodiment of this application;

[0052] FIG. 5 is a diagram of a structure of still another beamforming apparatus according to an embodiment of this application;

[0053] FIG. 6 is a diagram of a structure of still another beamforming apparatus according to an embodiment of this application;

[0054] FIG. 7 is a diagram of a structure of still another beamforming apparatus according to an embodiment of this application;

[0055] FIG. 8 is a diagram of a structure of still another beamforming apparatus according to an embodiment of this application;

[0056] FIG. 9 is a diagram of a structure of a beamforming apparatus in a heterodyne coherent detection scenario according to an embodiment of this application;

[0057] FIG. 10A is a diagram of a structure of another beamforming apparatus in a heterodyne coherent detection scenario according to an embodiment of this application;

[0058] FIG. 10B is a diagram of a structure of still another beamforming apparatus in a heterodyne coherent detection scenario according to an embodiment of this application;

[0059] FIG. 11 is a diagram of a structure of still another beamforming apparatus in a heterodyne coherent detection scenario according to an embodiment of this application;

[0060] FIG. 12 is a diagram of a structure of a beamforming apparatus in a homodyne coherent detection scenario according to an embodiment of this application;

[0061] FIG. 13 is a diagram of a structure of still another beamforming apparatus according to an embodiment of this application;

[0062] FIG. 14 is a diagram of a structure of still another beamforming apparatus according to an embodiment of this application;

[0063] FIG. 15 is a diagram of a structure of still another beamforming apparatus according to an embodiment of this application;

[0064] FIG. 16 is a diagram of a structure of still another beamforming apparatus according to an embodiment of this application;

[0065] FIG. 17 is a diagram of a structure of still another beamforming apparatus according to an

embodiment of this application;

[0066] FIG. **18** is a diagram of a structure of still another beamforming apparatus according to an embodiment of this application;

[0067] FIG. **19** is a diagram of a structure of still another beamforming apparatus according to an embodiment of this application;

[0068] FIG. **20** is a schematic flowchart of a beamforming method according to an embodiment of this application; and

[0069] FIG. **21** is a schematic flowchart of another beamforming method according to an embodiment of this application.

DESCRIPTION OF EMBODIMENTS

[0070] To make objectives, technical solutions, and advantages of embodiments of this application clearer, the following clearly describes the technical solutions in embodiments of this application with reference to the accompanying drawings in embodiments of this application. Clearly, the described embodiments are merely a part rather than all of embodiments of this application. All other embodiments obtained by persons of ordinary skill in the art based on embodiments of this application without creative efforts shall fall within the protection scope of this application.

[0071] The term “at least one” in this application means one or more, that is, includes one, two, three, or more; and the term “a plurality of” means two or more, that is, includes two, three, or more. In addition, it should be understood that in descriptions of this application, the terms such as “first” and “second” are merely used for distinguishing and description, but should not be understood as indicating or implying relative importance, or should not be understood as indicating or implying a sequence. “And/or” describes an association relationship between associated objects and represents that three relationships may exist. For example, A and/or B may represent the following cases: Only A exists, both A and B exist, and only B exists, where A and B may be singular or plural. The character “/” generally indicates an “or” relationship between associated objects. “At least one of the following items (pieces)” or a similar expression thereof refers to any combination of these items, including any combination of singular items (pieces) or plural items (pieces). For example, at least one of a, b, or c may indicate: a, b, c, a-b, a-c, b-c, or a-b-c, where a, b, and c may be singular or plural.

[0072] Beamforming is also referred to as beam shaping and spatial filtering, is a signal processing technology for array-based directional signal sending and receiving, and can be used at both a signal transmitting end and a signal receiving end. In the beamforming technology, a phase or delay of each transmitted signal in an array antenna is controlled, so that beams are interfered with and combined in a specific wavefront direction, and therefore a direction pointing angle of a signal beam is changed.

[0073] A current mainstream technical solution of beamforming is to use an electrical device to perform phase control, to implement a beamforming function. The beamforming solution using the electrical device cannot meet a requirement of increased antenna data caused by expansion of signal capacity. Therefore, currently, it is provided to expand a traffic channel by using characteristics of polarization and mode multiplexing of light. In a possible beamforming solution for light, a tunable laser device is used as a light source, a plurality of Bragg gratings with different center wavelengths are written onto a waveguide optical delay line, and Bragg gratings with a same center wavelength are enabled to locate at different locations in adjacent waveguide optical delay lines, to control a transmission delay of an optical signal by selecting an output wavelength of the laser device. In this way, a beamforming effect is achieved for a microwave signal, as shown in FIG. **1**.

[0074] Embodiments of this application provide a beamforming apparatus and method, to implement a fully connected beamforming solution by using a wavelength division multiplexing technology. The method and the apparatus are based on a same inventive concept. Because the method and the apparatus have a similar problem-resolving principle, for implementations of the

apparatus and the method, refer to each other, and no repeated description is provided.

[0075] FIG. 2A shows a beamforming apparatus according to an embodiment of this application. An example in which M radio frequency signals are transmitted through N antenna ends is used. [0076] With reference to FIG. 2A, the beamforming apparatus includes a first laser signal source **210**, M modulators **220**, a first optical splitting device **230**, N wavelength selection units **240**, and N*M phase shifters **250**. For ease of distinguishing between the N wavelength selection units **240**, in FIG. 2A, the N wavelength selection units **240** are referred to as a wavelength selection unit **240-1** to a wavelength selection unit **240-N**. The first laser signal source **210** is separately coupled to the M modulators **220**. Coupling refers to optical connection between inputs and outputs of two or more optical devices, where light energy is transmitted from one side to another side through interaction. For example, two optical devices may be connected through an optical fiber. The M modulators **220** are separately coupled to the first optical splitting device **230**. It may be understood that one end of each of the M modulators **220** is coupled to the first laser signal source **210**, and the other end of each modulator is coupled to the first optical splitting device **230**. The first optical splitting device **230** is separately coupled to the N wavelength selection units **240**. Every M phase shifters **250** in the N*M phase shifters **250** are coupled to one wavelength selection unit **240**. It may be understood that the N*M phase shifters **250** are divided into N phase shifter groups. In an example, in FIG. 2A, the N phase shifter groups are referred to as a phase shifter group 1 to a phase shifter group N. Each group includes M phase shifters **250**. For ease of distinguishing, M phase shifters **250** included in the phase shifter group 1 are referred to as a phase shifter **250-1-1** to a phase shifter **250-1-M**; M phase shifters **250** included in a phase shifter group 2 are referred to as a phase shifter **250-2-1** to a phase shifter **250-2-M**; and by analogy, M phase shifters **250** included in the phase shifter group N are referred to as a phase shifter **250-N-1** to a phase shifter **250-N-M**. The N phase shifter groups are in a one-to-one correspondence with the N wavelength selection units **240**, and M phase shifters included in one phase shifter group are connected to a corresponding wavelength selection unit **240**. The phase shifter group 1 is used as an example. The phase shifter **250-1-1** to the phase shifter **250-1-M** in the phase shifter group 1 are coupled to the wavelength selection unit **240-1**.

[0077] In a possible implementation, the beamforming apparatus further includes a detector array and a radio frequency antenna array, as shown in FIG. 2B. For example, the detector array includes N detectors. For example, the radio frequency antenna array includes N radio frequency antenna units **270**. For ease of distinguishing between the M modulators **220**, in FIG. 2B, the M modulators **220** are referred to as a modulator **220-1** to a modulator **220-M**. For ease of distinguishing, in FIG. 2B, the N detectors **260** are referred to as a detector **260-1** to a detector **260-N**. For ease of distinguishing, in FIG. 2B, the N radio frequency antenna units **270** are referred to as a radio frequency antenna unit **270-1** to a radio frequency antenna unit **270-N**. Ends of the N detectors **260** are coupled to the N wavelength selection units **240** in a one-to-one correspondence, and the other ends of the N detectors **260** are coupled to the N radio frequency antenna units **270** in a one-to-one correspondence.

[0078] The following describes a signal flow direction of the beamforming apparatus. FIG. 3 is a diagram of a signal flow of the beamforming apparatus.

[0079] The first laser signal source **210** transmits M laser signals with different wavelengths. The M laser signals with the different wavelengths respectively correspond to wavelengths $\lambda_{\text{sub.1}}$ to $\lambda_{\text{sub.M}}$. The M laser signals with the different wavelengths respectively enter different modulators **220**. For example, the M different laser signals are a laser signal **1** to a laser signal M. The laser signal **1** enters the modulator **220-1**; a laser signal **2** enters a modulator **220-2**; and by analogy, the laser signal M enters the modulator **220-M**.

[0080] The M modulators **220-1** to **220-M** are in a one-to-one correspondence with the laser signal **1** to the laser signal M, and respectively receive the laser signals of the corresponding wavelengths. Each modulator **220** modulates a radio frequency signal on a corresponding received laser signal,

to obtain M first modulated optical signals. Radio frequency signals modulated on any two of the M laser signals may be the same or different. A specific radio frequency signal that is modulated may be determined based on a user requirement. The first modulated optical signals that are respectively obtained through modulation by the M modulators **220-1** to **220-M** enter the first optical splitting device **230**.

[0081] After receiving the M first modulated optical signals, the first optical splitting device **230** couples the M first modulated optical signals, divides one path of coupled optical signals into N paths of optical signals, and distributes the N paths of optical signals to N optical paths, to obtain N second modulated optical signals corresponding to the N optical paths. The first optical splitting device **230** sends the N second modulated optical signals to the N wavelength selection units **240** in a one-to-one correspondence. After receiving the second modulated optical signal, each wavelength selection unit **240** sends, in a one-to-one correspondence, optical signals with M wavelengths included in the second modulated optical signal to M phase shifters coupled to the wavelength selection unit **240**.

[0082] With reference to FIG. 3, the wavelength selection unit **240-1** is used as an example. After receiving the second modulated optical signal, the wavelength selection unit **240-1** sends, in a one-to-one correspondence, optical signals with the M wavelengths in the second modulated optical signal to the phase shifter **250-1-1** to the phase shifter **250-1-M**. For example, the phase shifter **250-1-1** corresponds to an optical signal whose wavelength is $\lambda_{\text{sub.1}}$; a phase shifter **250-1-2** corresponds to an optical signal whose wavelength is $\lambda_{\text{sub.2}}$; and by analogy, the phase shifter **250-1-M** corresponds to an optical signal whose wavelength is $\lambda_{\text{sub.M}}$. Each phase shifter is configured to: adjust a phase of a received optical signal with a corresponding wavelength, and return the phase-adjusted optical signal to the wavelength selection unit. For example, the wavelength selection unit **240-1** extracts the optical signal whose wavelength is $\lambda_{\text{sub.1}}$ from the second modulated optical signal, and sends the optical signal to the phase shifter **250-1-1**. The phase shifter **250-1-1** adjusts a phase of the optical signal whose wavelength is $\lambda_{\text{sub.1}}$, and then sends the adjusted optical signal whose wavelength is $\lambda_{\text{sub.1}}$ to the wavelength selection unit **240-1**. The wavelength selection unit **240-1** extracts the optical signal whose wavelength is $\lambda_{\text{sub.2}}$ from the second modulated optical signal, and sends the optical signal to the phase shifter **250-1-2**. The phase shifter **250-1-2** adjusts a phase of the optical signal whose wavelength is $\lambda_{\text{sub.2}}$, and then sends the adjusted optical signal whose wavelength is $\lambda_{\text{sub.2}}$ to the wavelength selection unit **240-1**. By analogy, the wavelength selection unit **240-1** extracts the optical signal whose wavelength is $\lambda_{\text{sub.M}}$ from the second modulated optical signal, and sends the optical signal to the phase shifter **250-1-M**. The phase shifter **250-1-M** adjusts a phase of the optical signal whose wavelength is $\lambda_{\text{sub.M}}$, and then sends the adjusted optical signal whose wavelength is $\lambda_{\text{sub.M}}$ to the wavelength selection unit **240-1**. The N wavelength selection units **240** have a same function, and a manner of processing second modulated optical signals by other wavelength selection units **240-2** to **240-N** is similar to the manner of processing the second modulated optical signal by the wavelength selection unit **240-1**. Details are not described herein again.

[0083] In this embodiment of this application, phase shifters corresponding to a same wavelength (for example, $\lambda_{\text{sub.1}}$) in the N paths only adjust phases of optical signals whose wavelengths are $\lambda_{\text{sub.1}}$ and that are obtained after the N second modulated optical signals are split, and it does not indicate that the phase shifters corresponding to the wavelength $\lambda_{\text{sub.1}}$ in the N paths have a same phase shift variation amounts. Phase shift variation amounts of N phase shifters with a same wavelength in the N paths may be the same or may be different. This is not specifically limited in this embodiment of this application. For example, the phase shifter **250-1-1**, the phase shifter **250-2-1**, . . . , and the phase shifter **250-N-1** are all configured to perform phase shifting on optical signals whose wavelengths are $\lambda_{\text{sub.1}}$, and phase shift variation amounts caused by the phase shifter **250-1-1**, the phase shifter **250-2-1**, . . . , and the phase shifter **250-N-1** to the optical signals whose wavelengths are $\lambda_{\text{sub.1}}$ may be the same or different. Similarly, the phase shifter **250-1-2**, a

phase shifter **250-2-2**, . . . , and a phase shifter **250-N-2** are all configured to perform phase shifting on optical signals whose wavelengths are $\lambda_{\text{sub.2}}$, and phase shift variation amounts caused by the phase shifter **250-1-2**, the phase shifter **250-2-2**, . . . , and the phase shifter **250-N-2** to the optical signals whose wavelengths are $\lambda_{\text{sub.2}}$ may be the same or different. The rest may be deduced by analogy.

[0084] Each wavelength selection unit **240** sends, to a corresponding first detector **260**, a phase-shifted optical signal received from each phase shifter **250**. In some embodiments, each wavelength selection unit **240** fuses phase-shifted optical signals received from the M phase shifters **250**, to obtain a third modulated optical signal, and sends the third modulated optical signal to the corresponding first detector **260**. For example, as shown in FIG. 3, after receiving phase-shifted optical signals sent by the phase shifter **250-1-1** to the phase shifter **250-1-M**, the wavelength selection unit **240-1** fuses the M phase-shifted optical signals, and sends a fused optical signal to the detector **260-1**. Further, the detector **260-1** is configured to perform optical-to-electrical conversion on the received third modulated optical signal to obtain an electrical signal. Then, the radio frequency antenna unit **270-1** coupled to the detector **260-1** sends the electrical signal. Similarly, after receiving phase-shifted optical signals sent by the phase shifter **250-2-1** to the phase shifter **250-2-M**, the wavelength selection unit **240-2** fuses the M phase-shifted optical signals, and sends a fused optical signal to a detector **260-2**. Further, the detector **260-2** is configured to perform optical-to-electrical conversion on the received third modulated optical signal to obtain an electrical signal. Then, a radio frequency antenna unit **270-2** coupled to the detector **260-2** sends the electrical signal. The rest may be deduced by analogy.

[0085] The following describes the devices mentioned above. For ease of description subsequently, illustrations of the devices are not provided again. [0086] (1) First laser signal source **210**:

[0087] In a possible example, the first laser signal source **210** may include M laser devices, as shown in FIG. 4. In FIG. 4, the laser device included in the first laser signal source is referred to as a first laser device. The M laser devices are a first laser device **1** to a first laser device M. Wavelengths of laser signals sent by the first laser device **1** to the first laser device M are different. The wavelengths of the laser signals respectively transmitted by the first laser device **1** to the first laser device M are $\lambda_{\text{sub.1}}$ to $\lambda_{\text{sub.M}}$.

[0088] Powers of the laser signals generated by the M first laser devices may be the same or may be different. A wavelength spacing between laser signals transmitted by any two first laser devices in the first laser device **1** to the first laser device M may be determined based on an actual requirement. For example, the wavelength spacing between the laser signals transmitted by the any two first laser devices is greater than a specified threshold.

[0089] In another possible example, the first laser signal source **210** uses a multi-wavelength laser device. With reference to FIG. 5, the first laser signal source **210** includes a multi-wavelength laser device **211** and a wavelength division multiplexer **212**. The multi-wavelength laser device **211** may transmit one path of optical signals with the M wavelengths. The wavelength division multiplexer **212** distributes the path of optical signals with the M wavelengths into M paths of optical signals. Wavelengths of the M paths of optical signals are different. [0090] (2) Modulator **220**: In this embodiment of this application, there is no specific limitation on a modulation format of the modulator **220**. For example, a coherent modulation scheme or an intensity modulation scheme may be used. [0091] (3) First optical splitting device **230**:

[0092] In a possible example, with reference to FIG. 6, the first optical splitting device **230** may include a wavelength division multiplexer **231** and a beam splitter **232**. A function of the wavelength division multiplexer **231** is to couple, into one path of signal, M optical signals that are with different wavelengths, that are obtained by modulating radio frequency signals, and that are sent by the M modulators **220**. For example, the wavelength division multiplexer **231** is connected to the beam splitter **232** through a waveguide. The wavelength division multiplexer **231** couples, into a same waveguide, the M optical signals that are with the different wavelengths, that are

obtained by modulating the radio frequency signals, and that are sent by the M modulators **220**. A function of the beam splitter **232** is to split one path of input light transmitted by the wavelength division multiplexer **231** into N paths of optical signals (wavelengths are not distinguished) with a same power, where each light includes a plurality of optical signals with different wavelengths; and separately transmit the obtained N paths of optical signals to the N wavelength selection units **240** through waveguides. For example, each path of light may include optical signals whose wavelengths are $\lambda_{\text{sub.1}}$, $\lambda_{\text{sub.2}}$, . . . , and $\lambda_{\text{sub.M}}$. A value of N may be determined based on a quantity of wavelength selection units **240**, and N is an integer greater than or equal to 2. Because the M optical signals that are with the different wavelengths, that are obtained by modulating the radio frequency signals, and that are sent by the M modulators **220** respectively carry the radio frequency signals, in this manner, the N paths of optical signals obtained through the wavelength division multiplexer **231** and the beam splitter **232** each include radio frequency signals carried by a plurality of optical signals whose wavelengths are $\lambda_{\text{sub.1}}$, $\lambda_{\text{sub.2}}$, . . . , and $\lambda_{\text{sub.M}}$. For example, when the M wavelength selection units **240** are connected to in the wavelength division multiplexing manner shown in FIG. 6, devices in the entire beamforming apparatus may be connected through waveguides.

[0093] The first optical splitting device **230** may be a planar waveguide. In this case, the M optical signals that are with the different wavelengths, that are obtained by modulating the radio frequency signals, and that are sent by the M modulators **220** are transmitted to the M wavelength selection units **240** through planar waveguide space. [0094] (4) Wavelength selection unit **240**:

[0095] In some embodiments, the wavelength selection unit may be a reconfigurable optical add-drop multiplexer (ROADM), or another reconfigurable optical device that can be used for wavelength selection. [0096] (5) Phase shifter **250**: In this embodiment of this application, no specific limitation is imposed on the phase shifter, and all optical devices with a phase shifting function of adjusting a phase of an optical signal are applicable to this application. [0097] (6) Radio frequency antenna array:

[0098] The radio frequency antenna array may include the N radio frequency antenna units **270**. In some possible examples, with reference to FIG. 7, the radio frequency antenna unit **270** may include a front-end amplifier **271** and an antenna element **272**. In another possible example, the radio frequency antenna unit **270** may include an antenna element. The front-end amplifier **271** is configured to: perform front-end amplification on an electrical signal from the detector **260**, and send the electrical signal to the antenna element **272**. The antenna element **272** transmits the received electrical signal from the front-end amplifier **271**.

[0099] In a possible implementation, the beamforming apparatus may further include a control unit **280**. With reference to FIG. 8, the control unit **280** may be configured to control phase shift spacings of the N*M phase shifters. For example, the control unit **280** may control phase shift variation amounts of N phase shifters of a same wavelength in the N paths to be the same or different. For example, the control unit **280** controls the phase shift variation amounts caused by the phase shifter **250-1-1**, the phase shifter **250-2-1**, . . . , and the phase shifter **250-N-1** to the optical signals whose wavelengths are $\lambda_{\text{sub.1}}$ to be the same or different. The control unit **280** controls the phase shift variation amounts caused by the phase shifter **250-1-2**, the phase shifter **250-2-2**, . . . , and the phase shifter **250-N-2** to the optical signals whose wavelengths are $\lambda_{\text{sub.2}}$ to be the same or different. The rest may be deduced by analogy.

[0100] The control unit **280** provided in the foregoing embodiment of this application may be a central processing unit (CPU), or may be another general-purpose processor, a digital signal processor (DSP), an application-specific integrated circuit (ASIC), a field programmable gate array (FPGA) or another programmable logic device, a discrete gate or transistor logic device, a discrete hardware component, or the like. The general-purpose processor may be a microprocessor, or the processor may be any conventional processor or the like. It should be understood that the control unit **280** in the foregoing embodiment of this application may be implemented by one or more

processors. A quantity of processors may be adjusted based on an actual application scenario. This is merely an example for description herein, and is not limited.

[0101] In this embodiment of this application, the control unit **280** controls the phase shift variation amounts caused by the phase shifter **250-1-1**, the phase shifter **250-2-1**, . . . , and the phase shifter **250-N-1** to the optical signals whose wavelengths are $\lambda_{\text{sub.1}}$, that is, may change a transmit angle of a radio frequency signal **1** at the antenna end after beamforming. The control unit **280** controls the phase shift variation amounts caused by the phase shifter **250-1-2**, the phase shifter **250-2-2**, . . . , and the phase shifter **250-N-2** to the optical signals whose wavelengths are $\lambda_{\text{sub.2}}$, that is, may change a transmit angle of a radio frequency signal **2** at the antenna end after beamforming. By analogy, the control unit **280** may change transmit angles of radio frequency signals **1** to **M** at the antenna ends after beamforming, to complete beamforming. In this embodiment of this application, a quantity **M** of radio frequency signals that can be simultaneously transmitted is not specifically limited. The **M** radio frequency signals may be transmitted in a time division multiplexing manner, or may be all simultaneously transmitted, to implement full connection of the **M** radio frequency signals.

[0102] In a possible implementation, the beamforming apparatus may further include a memory, which is not shown in the figure. The memory may be configured to store program instructions. The control unit **280** may be configured to read a program instruction from the memory to control the phase shifter. In some scenarios, the memory may alternatively be deployed inside the control unit **280**. For example, the control unit **280** includes a processor and the memory. In other words, the processor and the memory may be integrated together, or the memory and the processor may be connected through an interface, which may be adjusted based on the actual application scenario. This is not specifically limited in this embodiment of this application.

[0103] There may be one or more memories in this embodiment of this application. A quantity of memories may be adjusted based on the actual application scenario. This is merely an example for description herein, and is not limited.

[0104] It should be further understood that, in this embodiment of this application, the memory, the readable storage medium, or the like described in the foregoing embodiment may be a volatile memory or a non-volatile memory, or may include both a volatile memory and a non-volatile memory. The non-volatile memory may be a read-only memory (ROM), a programmable read-only memory (PROM), an erasable programmable read-only memory (EPROM), an electrically erasable programmable read-only memory (EEPROM), or a flash memory. The volatile memory may be a random access memory (RAM), which is used as an external cache. By way of example, and not limitation, RAMs in many forms may be used, for example, a static random access memory (SRAM), a dynamic random access memory (DRAM), a synchronous dynamic random access memory (SDRAM), a double data rate synchronous dynamic random access memory (DDR SDRAM), an enhanced synchronous dynamic random access memory (ESDRAM), a synchlink dynamic random access memory (SLDRAM), and a direct rambus random access memory (DRRAM).

[0105] This embodiment of this application may be applied to an application scenario of direct modulation and direct detection, or may be applied to an application scenario of heterodyne or homodyne detection. For example, the beamforming apparatus described in the embodiment corresponding to FIG. 2A to FIG. 7 may be applied to the application scenario of direct modulation and direct detection.

[0106] In a possible implementation, when the beamforming apparatus is applied to an application scenario of heterodyne coherent detection, the beamforming apparatus may further include a second laser signal source **310** and a second optical splitting device **320**, as shown in FIG. 9. The second laser signal source **310** is configured to transmit **M** carrier optical signals that are in a one-to-one correspondence with the **M** laser signals. Wavelengths of the **M** carrier optical signals are the same as the wavelengths of the corresponding laser signals. The second optical splitting device

320 is configured to couple and distribute the M carrier optical signals to the N detectors **260**. Then, each of the N detectors **260** performs coherent detection on a received third modulated optical signal based on received carrier optical signals, and performs optical-to-electrical conversion on the coherently detected third modulated optical signal to obtain an electrical signal. [0107] Carrier optical signals **1** to M are coupled into one path of optical signals through the second optical splitting device **320**, and then are split into N paths of optical signals. Each of the N paths of optical signals may include optical signals with the M wavelengths. In the detector **260**, optical signals that have a same wavelength and that are separately in the received third modulated optical signal and the optical signals with the M wavelengths exhibit a beat frequency. Coherent detection and optical-to-electrical conversion are performed to obtain an electrical signal, and then the electrical signal is transmitted through the radio frequency antenna unit **270**.

[0108] A structure of the second laser signal source **310** is similar to that of the first laser signal source **210**. In a possible example, the second laser signal source **310** includes M laser devices, as shown in FIG. **10A**. In FIG. **10A**, the laser device included in the second laser signal source is referred to as a second laser device. The M laser devices are a second laser device **1** to a second laser device M. Wavelengths of carrier optical signals sent by the second laser device **1** to the second laser device M are different. The wavelengths of the carrier optical signals respectively transmitted by the second laser device **1** to the second laser device M are $\lambda_{\text{sub.1}}$ to $\lambda_{\text{sub.M}}$. Powers of the carrier optical signals generated by the M second laser devices may be the same or may be different. A wavelength spacing between laser signals transmitted by any two second laser devices in the second laser device **1** to the second laser device M may be determined based on the actual requirement. For example, the wavelength spacing between the laser signals transmitted by the any two second laser devices is greater than a specified threshold.

[0109] In another possible example, the second laser signal source **310** uses a multi-wavelength laser device. With reference to FIG. **10B**, the second laser signal source **310** includes a multi-wavelength laser device **311** and a wavelength division multiplexer **312**. The multi-wavelength laser device **311** may transmit one path of optical signals with the M wavelengths. The wavelength division multiplexer **312** distributes the path of optical signals with the M wavelengths into M paths of optical signals. Wavelengths of the M paths of optical signals are different, and are $\lambda_{\text{sub.1}}$ to $\lambda_{\text{sub.M}}$.

[0110] A structure of the second optical splitting device **320** is similar to that of the first optical splitting device **230**. In an example, with reference to FIG. **11**, the second optical splitting device **320** includes a wavelength division multiplexer **321** and a beam splitter **322**. A function of the wavelength division multiplexer **321** is to couple, into one path of signal, the M carrier optical signals that are with different wavelengths and that are sent by the second laser signal source **310**. For example, the wavelength division multiplexer **321** is connected to the beam splitter **322** through a waveguide. The wavelength division multiplexer **321** couples, into a same waveguide, the M carrier optical signals that are with the different wavelengths and that are sent by the second laser signal source **310**. A function of the beam splitter **322** is to split one path of input light transmitted by the wavelength division multiplexer **321** into N paths of optical signals (wavelengths are not distinguished) with a same power, where each light includes a plurality of optical signals with different wavelengths; and separately transmit the obtained N paths of optical signals to the N detectors **260** through waveguides.

[0111] In some embodiments, the second laser signal source **310** and the second optical splitting device **320** may also be deployed outside the beamforming apparatus as an independent device. When the independent device is connected to the beamforming apparatus, heterodyne coherent detection may be implemented. When the independent device is not connected to the beamforming apparatus, direct modulation and direct detection may be implemented.

[0112] In a possible implementation, when the beamforming apparatus is applied to an application scenario of homodyne coherent detection, the beamforming apparatus may further include a third

optical splitting device **410**, as shown in FIG. **12**. One end of the third optical splitting device **410** is coupled to the first laser signal source, the other end of the third optical splitting device **410** is coupled to the N detectors, and the third optical splitting device **410** is configured to couple and distribute the M laser signals to the N detectors **260**. Then, each of the N detectors **260** performs coherent detection on a received third modulated optical signal based on received laser signals, performs optical-to-electrical conversion on the coherently detected third modulated optical signal to obtain an electrical signal, and transmits the electrical signal through the radio frequency antenna unit **270**.

[0113] A structure of the third optical splitting device **410** is similar to those of the second optical splitting device **320** and the first optical splitting device **230**. For example, the third optical splitting device **410** also includes a wavelength division multiplexer and a beam splitter.

[0114] In some embodiments, the third optical splitting device **410** may also be deployed outside the beamforming apparatus as an independent device. When the independent device is connected to the beamforming apparatus, homodyne coherent detection may be implemented. When the independent device is not connected to the beamforming apparatus, direct modulation and direct detection may be implemented.

[0115] FIG. **13** shows another beamforming apparatus according to an embodiment of this application. An example in which M radio frequency signals are transmitted through N antenna ends is used. Different from the beamforming apparatus shown in FIG. **2A** to FIG. **12**, the beamforming apparatus does not need a wavelength selection unit. In this way, a quantity of devices of the beamforming apparatus can be reduced, costs can be reduced, and space utilization can be improved.

[0116] With reference to FIG. **13**, the beamforming apparatus includes a laser signal source **1310**, M modulators **1320**, M beam splitters **1330**, N wavelength division multiplexers **1350**, N*M phase shifters **1340**, N detectors **1360**, and N radio frequency antenna units **1370**. The N detectors **1360** may be arranged in an array manner. The N radio frequency antenna units may be arranged in an array manner.

[0117] The laser signal source **1310** is coupled to the M modulators **1320**. The M modulators **1320** are coupled to the M beam splitters **1330** in a one-to-one correspondence. Each beam splitter **1330** is coupled to N phase shifters **1340**. Groups of N phase shifters **1340** coupled to different beam splitters **1330** are different. For ease of distinguishing, in FIG. **13**, the M modulators **1320** are referred to as a modulator **1320-1** to a modulator **1320-M**. The M beam splitters **1330** are referred to as a beam splitter **1330-1** to a beam splitter **1330-M**.

[0118] Every M phase shifters **1340** in the N*M phase shifters **1340** may be considered as a phase shifter group. M phase shifters **1340** included in one phase shifter group are all connected to a same wavelength division multiplexer **1350**. For example, an input end of a wavelength division multiplexer **1350** includes M ports, and the M ports of the input end of the wavelength division multiplexer **1350** are connected to output ends of M phase shifters **1340** included in a phase shifter group. Input ends of M phase shifters **1340** included in a phase shifter group are respectively connected to different beam splitters **1330**. In other words, an input end of each of the N wavelength division multiplexers **1350** is coupled to output ends of M phase shifters **1340**, and input ends of M phase shifters **1340** coupled to a same wavelength division multiplexer **1350** are respectively coupled to different beam splitters **1330**.

[0119] In an example, the N phase shifter groups are referred to as a phase shifter group 1 to a phase shifter group N. Each group includes M phase shifters **1340**. For ease of distinguishing, M phase shifters **1340** included in the phase shifter group 1 are referred to as a phase shifter **1340-1-1** to a phase shifter **1340-1-M**; M phase shifters **1340** included in a phase shifter group 2 are referred to as a phase shifter **1340-2-1** to a phase shifter **1340-2-M**; and by analogy, M phase shifters **1340** included in the phase shifter group N are referred to as a phase shifter **1340-N-1** to a phase shifter **1340-N-M**. In FIG. **13**, the phase shifter **1340-1-1** to the phase shifter **1340-1-M** in the phase shifter

group 1 are coupled to an input end of a wavelength division multiplexer **1350-1**; the phase shifter **1340-2-1** to the phase shifter **1340-2-M** in the phase shifter group 2 are coupled to an input end of a wavelength division multiplexer **1350-2**; and by analogy, the phase shifter **1340-N-1** to the phase shifter **1340-N-M** in the phase shifter group N are coupled to an input end of a wavelength division multiplexer **1350-N**. Therefore, it may be understood that each beam splitter **1330** is connected to N phase shifters. The beam splitter **1330-1** is connected to the phase shifter **1340-1-1** to the phase shifter **1340-N-1**; a beam splitter **1330-2** is connected to a phase shifter **1340-1-2** to a phase shifter **1340-N-2**; and by analogy, the beam splitter **1330-M** is connected to the phase shifter **1340-1-M** to the phase shifter **1340-N-M**.

[0120] Output ends of the N wavelength division multiplexers **1350** are coupled to input ends of the N detectors **1360** in a one-to-one correspondence. Output ends of the N detectors **1360** are coupled to input ends of the N radio frequency antenna units **1370** in a one-to-one correspondence.

[0121] The following describes a signal flow direction of the beamforming apparatus. FIG. 14 is a diagram of a signal flow of the beamforming apparatus.

[0122] The laser signal source **1310** transmits M laser signals with different wavelengths. The M laser signals with the different wavelengths respectively correspond to wavelengths $\lambda_{\text{sub.1}}$ to $\lambda_{\text{sub.M}}$. The M laser signals with the different wavelengths respectively enter different modulators **1320**. For example, the M different laser signals are a laser signal **1** to a laser signal M. The laser signal **1** enters the modulator **1320-1**; a laser signal **2** enters a modulator **1320-2**; and by analogy, the laser signal M enters the modulator **1320-M**. The M modulators **1320-1** to **1320-M** are in a one-to-one correspondence with the laser signal **1** to the laser signal M, and respectively receive the laser signals of the corresponding wavelengths. Each modulator **1320** modulates a radio frequency signal on a corresponding received laser signal, to obtain M first modulated optical signals. Radio frequency signals modulated on any two of the M laser signals may be the same or different. A specific radio frequency signal that is modulated may be determined based on a user requirement. The first modulated optical signals that are respectively obtained through modulation by the M modulators **1320-1** to **1320-M** enter different beam splitters **1330**. A first modulated optical signal obtained through modulation by the modulator **1320-1** enters the beam splitter **1330-1**; a first modulated optical signal obtained through modulation by the modulator **1320-2** enters the beam splitter **1330-2**; and by analogy, a first modulated optical signal obtained through modulation by the modulator **1320-M** enters the beam splitter **1330-M**. After receiving the first modulated optical signal, each beam splitter **1330** splits the received first modulated optical signal into N second modulated optical signals, and distributes the N second modulated optical signals to N phase shifters connected to an output end. For example, the beam splitter **1330-1** respectively distributes N second modulated optical signals to the phase shifter **1340-1-1** to the phase shifter **1340-N-1**; the beam splitter **1330-2** respectively distributes N second modulated optical signals to the phase shifter **1340-1-2** to the phase shifter **1340-N-2**; and by analogy, the beam splitter **1330-M** respectively distributes N second modulated optical signals to the phase shifter **1340-1-M** to the phase shifter **1340-N-M**.

[0123] Each phase shifter **1340** performs phase adjustment on the received second modulated optical signal, and sends a third modulated optical signal obtained through adjustment to a wavelength division multiplexer **1350** coupled to the phase shifter. The phase shifter **1340-1-1** to the phase shifter **1340-1-M** are coupled to the input end of the wavelength division multiplexer **1350-1**, and the phase shifter **1340-1-1** to the phase shifter **1340-1-M** separately send third modulated optical signals obtained through phase adjustment to the wavelength division multiplexer **1350-1**. The wavelength division multiplexer **1350-1** combines the M third modulated optical signals into a fourth modulated optical signal, and sends the fourth modulated optical signal to a detector **1360-1**. The detector **1360-1** performs optical-to-electrical conversion on the fourth modulated optical signal to obtain an electrical signal, and transmits the electrical signal through a radio frequency antenna unit **1370-1**. The phase shifter **1340-2-1** to the phase shifter **1340-2-M** are

coupled to the input end of the wavelength division multiplexer **1350-2**, and the phase shifter **1340-2-1** to the phase shifter **1340-2-M** separately send third modulated optical signals obtained through phase adjustment to the wavelength division multiplexer **1350-2**. The wavelength division multiplexer **1350-2** combines the M third modulated optical signals into a fourth modulated optical signal, and sends the fourth modulated optical signal to a detector **1360-2**. The detector **1360-2** performs optical-to-electrical conversion on the fourth modulated optical signal to obtain an electrical signal, and transmits the electrical signal through a radio frequency antenna unit **1370-2**. By analogy, the phase shifter **1340-N-1** to the phase shifter **1340-N-M** are coupled to the input end of the wavelength division multiplexer **1350-N**, and the phase shifter **1340-N-1** to the phase shifter **1340-N-M** separately send third modulated optical signals obtained through phase adjustment to the wavelength division multiplexer **1350-N**. The wavelength division multiplexer **1350-N** combines the M third modulated optical signals into a fourth modulated optical signal, and sends the fourth modulated optical signal to a detector **1360-N**. The detector **1360-N** performs optical-to-electrical conversion on the fourth modulated optical signal to obtain an electrical signal, and transmits the electrical signal through a radio frequency antenna unit **1370-N**.

[0124] In this embodiment of this application, phase shifters corresponding to a same wavelength (for example, $\lambda_{\text{sub.1}}$) in N paths, namely, phase shifters connected to a same beam splitter **1330**, only adjust phases of optical signals whose wavelengths are $\lambda_{\text{sub.1}}$ and that are obtained after the N second modulated optical signals are split, and it does not indicate that phase shifters corresponding to the wavelength $\lambda_{\text{sub.1}}$ in the N paths have a same phase variation amount. Phase shift variation amounts of N phase shifters with a same wavelength in the N paths may be the same or may be different. This is not specifically limited in this embodiment of this application. For example, the phase shifter **1340-1-1**, the phase shifter **1340-2-1**, . . . , and the phase shifter **1340-N-1** are all configured to perform phase shifting on optical signals whose wavelengths are $\lambda_{\text{sub.1}}$ and phase shift variation amounts caused by the phase shifter **1340-1-1**, the phase shifter **1340-2-1**, . . . , and the phase shifter **1340-N-1** to the optical signals whose wavelengths are $\lambda_{\text{sub.1}}$ may be the same or different. Similarly, the phase shifter **1340-1-2**, a phase shifter **1340-2-2**, . . . , and the phase shifter **1340-N-2** are all configured to perform phase shifting on optical signals whose wavelengths are $\lambda_{\text{sub.2}}$, and phase shift variation amounts caused by the phase shifter **1340-1-2**, the phase shifter **1340-2-2**, . . . , and the phase shifter **1340-N-2** to the optical signals whose wavelengths are $\lambda_{\text{sub.2}}$ may be the same or different. The rest may be deduced by analogy.

[0125] A structure of the laser signal source **1310** is similar to that of the first laser signal source **210**. Details are not described herein again. Refer to FIG. 15 and FIG. 16. In this embodiment of this application, there is no specific limitation on a modulation format of the modulator **1320**. For example, a coherent modulation scheme or an intensity modulation scheme may be used. In this embodiment of this application, no specific limitation is imposed on the phase shifter, and all optical devices with a phase shifting function of adjusting a phase of an optical signal are applicable to this application. Structures of the N radio frequency antenna units **1370** are similar to that of the radio frequency antenna **270**. Details are not described herein again.

[0126] In a possible implementation, the beamforming apparatus may further include a control unit **1380**. With reference to FIG. 17, the control unit **1380** may be configured to control phase shift spacings of the N*M phase shifters. For example, the control unit **1380** may control phase shift variation amounts of N phase shifters of a same wavelength in the N paths to be the same or different. For example, the control unit **1380** controls the phase shift variation amounts caused by the phase shifter **1340-1-1**, the phase shifter **1340-2-1**, . . . , and the phase shifter **1340-N-1** to the optical signals whose wavelengths are $\lambda_{\text{sub.1}}$ to be the same or different. The control unit **1380** controls the phase shift variation amounts caused by the phase shifter **1340-1-2**, the phase shifter **1340-2-2**, . . . , and the phase shifter **1340-N-2** to the optical signals whose wavelengths are $\lambda_{\text{sub.2}}$ to be the same or different. The rest may be deduced by analogy.

[0127] The control unit **1380** provided in the foregoing embodiment of this application may be a

central processing unit (CPU), or may be another general-purpose processor, a digital signal processor (DSP), an application-specific integrated circuit (ASIC), a field programmable gate array (FPGA) or another programmable logic device, a discrete gate or transistor logic device, a discrete hardware component, or the like. The general-purpose processor may be a microprocessor, or the processor may be any conventional processor or the like. It should be understood that the control unit **1380** in the foregoing embodiment of this application may be implemented by one or more processors. A quantity of processors may be adjusted based on an actual application scenario. This is merely an example for description herein, and is not limited.

[0128] In this embodiment of this application, the control unit **1380** controls the phase shift variation amounts caused by the phase shifter **1340-1-1**, the phase shifter **1340-2-1**, . . . , and the phase shifter **1340-N-1** to the optical signals whose wavelengths are $\lambda_{\text{sub.1}}$, that is, may change a transmit angle of a radio frequency signal **1** at the antenna end after beamforming. The control unit **1380** controls the phase shift variation amounts caused by the phase shifter **1340-1-2**, the phase shifter **1340-2-2**, . . . , and the phase shifter **1340-N-2** to the optical signals whose wavelengths are $\lambda_{\text{sub.2}}$, that is, may change a transmit angle of a radio frequency signal **2** at the antenna end after beamforming. By analogy, the control unit **1380** may change transmit angles of radio frequency signals **1** to **M** at the antenna ends after beamforming, to complete beamforming. In this embodiment of this application, a quantity **M** of radio frequency signals that can be simultaneously transmitted is not specifically limited. The **M** radio frequency signals may be transmitted in a time division multiplexing manner, or may be all simultaneously transmitted, to implement full connection of the **M** radio frequency signals.

[0129] In a possible implementation, the beamforming apparatus may further include a memory, which is not shown in the figure. The memory may be configured to store program instructions. The control unit **1380** may be configured to read a program instruction from the memory to control the phase shifter. In some scenarios, the memory may alternatively be deployed inside the control unit **1380**. For example, the control unit **1380** includes a processor and the memory. In other words, the processor and the memory may be integrated together, or the memory and the processor may be connected through an interface, which may be adjusted based on the actual application scenario. This is not specifically limited in this embodiment of this application.

[0130] This embodiment of this application may be applied to an application scenario of direct modulation and direct detection, or may be applied to an application scenario of heterodyne or homodyne detection. For example, the beamforming apparatus described in the embodiment corresponding to FIG. **8** to FIG. **15** may be applied to the application scenario of direct modulation and direct detection.

[0131] In a possible implementation, when the beamforming apparatus is applied to an application scenario of heterodyne coherent detection, the beamforming apparatus may further include a laser signal source **1410** and an optical splitting device **1420**, as shown in FIG. **18**. The laser signal source **1410** is configured to transmit **M** carrier optical signals that are in a one-to-one correspondence with the **M** laser signals. Wavelengths of the **M** carrier optical signals are the same as the wavelengths of the corresponding laser signals. The optical splitting device **1420** is configured to couple and distribute the **M** carrier optical signals to the **N** detectors **1360**. Then, each of the **N** detectors **1360** performs coherent detection on a received fourth modulated optical signal based on received carrier optical signals, and performs optical-to-electrical conversion on the coherently detected fourth modulated optical signal to obtain an electrical signal. A structure of the laser signal source **1410** is similar to that of the second laser signal source **310**. Details are not described herein again. A structure of the optical splitting device **1420** is similar to that of the second optical splitting device **320**. Details are not described herein again.

[0132] In some embodiments, the laser signal source **1410** and the optical splitting device **1420** may also be deployed outside the beamforming apparatus as an independent device. When the independent device is connected to the beamforming apparatus, heterodyne coherent detection may

be implemented. When the independent device is not connected to the beamforming apparatus, direct modulation and direct detection may be implemented.

[0133] In a possible implementation, when the beamforming apparatus is applied to an application scenario of homodyne coherent detection, the beamforming apparatus may further include an optical splitting device **1510**, as shown in FIG. **19**. One end of the optical splitting device **1510** is coupled to the laser signal source **1310**, the other end of the optical splitting device **1510** is coupled to the N detectors, and the optical splitting device **1510** is configured to couple and distribute the M laser signals to the N detectors **1360**. Then, each of the N detectors **1360** performs coherent detection on a received fourth modulated optical signal based on received laser signals, performs optical-to-electrical conversion on the coherently detected fourth modulated optical signal to obtain an electrical signal, and transmits the electrical signal through the radio frequency antenna unit **1370**.

[0134] In some embodiments, the optical splitting device **1510** may also be deployed outside the beamforming apparatus as an independent device. When the independent device is connected to the beamforming apparatus, homodyne coherent detection may be implemented. When the independent device is not connected to the beamforming apparatus, direct modulation and direct detection may be implemented.

[0135] There may be one or more memories in this embodiment of this application. A quantity of memories may be adjusted based on the actual application scenario. This is merely an example for description herein, and is not limited.

[0136] It should be further understood that, in this embodiment of this application, the memory, the readable storage medium, or the like described in the foregoing embodiment may be a volatile memory or a non-volatile memory, or may include both a volatile memory and a non-volatile memory. The non-volatile memory may be a read-only memory (ROM), a programmable read-only memory (PROM), an erasable programmable read-only memory (EPROM), an electrically erasable programmable read-only memory (EEPROM), or a flash memory. The volatile memory may be a random access memory (RAM), which is used as an external cache. By way of example, and not limitation, RAMs in many forms may be used, for example, a static random access memory (SRAM), a dynamic random access memory (DRAM), a synchronous dynamic random access memory (SDRAM), a double data rate synchronous dynamic random access memory (DDR SDRAM), an enhanced synchronous dynamic random access memory (ESDRAM), a synchlink dynamic random access memory (SLDRAM), and a direct rambus random access memory (DRRAM).

[0137] Based on the foregoing content and a same concept, this application provides a beamforming method. Refer to descriptions in FIG. **20**. The beamforming method may be applied to the beamforming apparatus shown in any one of the embodiments of FIG. **2A** to FIG. **12**. It may also be understood that the beamforming method may be implemented based on the beamforming apparatus shown in any one of the embodiments of FIG. **2A** to FIG. **12**. [0138] **2001**: Transmit M laser signals with different wavelengths.

[0139] For example, a first laser signal source **210** may send the M laser signals with the different wavelengths. For a structure and a specific implementation of the first laser signal source **210**, refer to the foregoing descriptions. Details are not described herein again. [0140] **2002**: Respectively modulate radio frequency signals on the M laser signals to obtain M first modulated optical signals, where radio frequency signals modulated on any two of the M laser signals are the same or different.

[0141] For example, M modulators **220** may respectively modulate the radio frequency signals on the M laser signals. [0142] **2003**: Combine and distribute the M first modulated optical signals into N optical paths, to obtain N second modulated optical signals corresponding to the N optical paths.

[0143] For example, a first optical splitting device **230** may combine and distribute the M first modulated optical signals into the N optical paths, to obtain the N second modulated optical signals

corresponding to the N optical paths. [0144] **2004**: On each optical path, perform phase shifting on optical signals with M wavelengths included in the second modulated optical signal, couple and perform optical-to-electrical conversion on the M phase-shifted optical signals to obtain an electrical signal, and transmit the electrical signal through an antenna.

[0145] For example, a wavelength selection unit **240** corresponding to each optical path may respectively select the optical signals of the M wavelengths from the second modulated optical signal and send the optical signals to corresponding phase shifters for phase shifting. Then, a detector **260** couples the M phase-shifted optical signals, performs optical-to-electrical conversion to obtain the electrical signal, and transmits the electrical signal through the antenna (or referred to as an antenna element). In some embodiments, the electrical signal may be further amplified by a front-end amplifier before being sent, and then transmitted through the antenna element.

[0146] In a possible implementation, the method may further include: transmitting M carrier optical signals that are in a one-to-one correspondence with the M laser signals; and combining and splitting the M carrier optical signals into N paths of optical signals, where each of the N paths of optical signals includes M carrier optical signals. Further, coupling and performing optical-to-electrical conversion on the M phase-shifted optical signals on each path to obtain the electrical signal includes: [0147] on each optical path, respectively performing coherent detection on the M phase-shifted optical signals based on the M received carrier optical signals, and performing optical-to-electrical conversion on the coherently detected optical signals to obtain the electrical signal.

[0148] In a possible implementation, the method may further include: coupling and splitting the M laser signals with the different wavelengths into N paths of optical signals, where each of the N paths of optical signals includes M laser signals with the different wavelengths.

[0149] Coupling and performing optical-to-electrical conversion on the M phase-shifted optical signals to obtain the electrical signal includes: [0150] respectively performing coherent detection on the M phase-shifted optical signals based on the M received laser signals, and performing optical-to-electrical conversion on the coherently detected optical signals to obtain the electrical signal.

[0151] Based on the foregoing content and a same concept, this application further provides another beamforming method. Refer to descriptions in FIG. **21**. The beamforming method may be applied to the beamforming apparatus shown in any one of the embodiments of FIG. **13** to FIG. **19**. It may also be understood that the beamforming method may be implemented based on the beamforming apparatus shown in any one of the embodiments of FIG. **13** to FIG. **19**. [0152] **2101**: Transmit M laser signals with different wavelengths.

[0153] For example, a laser signal source **1310** may send the M laser signals with the different wavelengths. For a structure and a specific implementation of the laser signal source **1310**, refer to the foregoing descriptions. Details are not described herein again. [0154] **2102**: Respectively modulate radio frequency signals on the M laser signals to obtain M first modulated optical signals, where radio frequency signals modulated on any two of the M laser signals are the same or different.

[0155] For example, M modulators **1320** may respectively modulate the radio frequency signals on the M laser signals. [0156] **2103**: Split each of the M first modulated optical signals into N second modulated optical signals, to obtain $N \times M$ second modulated optical signals.

[0157] For example, M beam splitters **1310** may respectively split the M first modulated optical signals, where each first modulated optical signal is split into N second modulated optical signals, to obtain the $N \times M$ second modulated optical signals. [0158] **2104**: Respectively perform phase adjustment on each of the $N \times M$ second modulated optical signals to obtain a third modulated optical signal, to obtain $N \times M$ third modulated optical signals.

[0159] For example, $N \times M$ phase shifters **1340** may process the $N \times M$ second modulated optical signals, to obtain the $N \times M$ third modulated optical signals. [0160] **2105**: Couple M third modulated

optical signals included in each of N third modulated optical signal groups in the N*M third modulated optical signals into a fourth modulated optical signal, to obtain N fourth modulated optical signals. Wavelengths of the M third modulated optical signals included in each of the N third modulated optical signal groups are different.

[0161] For example, N wavelength division multiplexers **1350** may couple the M third modulated optical signals included in each of the N third modulated optical signal groups in the N*M third modulated optical signals into the fourth modulated optical signal, to obtain the N fourth modulated optical signals. [0162] **2106**: Perform optical-to-electrical conversion on each of the N fourth modulated optical signals to obtain an electrical signal, and transmit the electrical signal through a corresponding antenna.

[0163] For example, N detectors **1360** may perform optical-to-electrical conversion on the N fourth modulated optical signals to obtain electrical signals, and transmit the electrical signals through antennas (or referred to as antenna elements). In some embodiments, the electrical signal may be further amplified by a front-end amplifier before being sent, and then transmitted through the antenna element.

[0164] Persons of ordinary skill in the art may understand that all or some of the steps performed by the control unit in the foregoing embodiments may be implemented by hardware or a program instructing related hardware. The program may be stored in a computer-readable storage medium. The storage medium mentioned above may be a read-only memory, a random access memory, or the like. Specifically, for example, the foregoing processing unit or processor may be a central processing unit, a general-purpose processor, a digital signal processor (DSP), an application-specific integrated circuit (ASIC), a field programmable gate array (FPGA) or another programmable logic device, a transistor logic device, a hardware component, or any combination thereof. Whether the functions are performed by hardware or software depends on particular applications and design constraints of the technical solutions. Persons skilled in the art may use different methods to implement the described functions for each particular application, but it should not be considered that the implementation goes beyond the scope of this application.

[0165] When software is used to implement the functions, all or some of the method steps described in the foregoing embodiments may be implemented in a form of a computer program product. The computer program product includes one or more computer instructions. When the computer program instructions are loaded and executed on a computer, the procedures or functions according to embodiments of this application are completely or partially generated. The computer may be a general-purpose computer, a dedicated computer, a computer network, or another programmable apparatus. The computer instructions may be stored in a computer-readable storage medium or may be transferred from a computer-readable storage medium to another computer-readable storage medium. For example, the computer instructions may be transferred from a website, computer, server, or data center to another website, computer, server, or data center in a wired (for example, a coaxial cable, an optical fiber, or a digital subscriber line (DSL)) or wireless (for example, infrared, radio, or microwave) manner. The computer-readable storage medium may be any usable medium accessible by a computer, or a data storage device, such as a server or a data center, integrating one or more usable media. The usable medium may be a magnetic medium (for example, a floppy disk, a hard disk, or a magnetic tape), an optical medium (for example, a BD), a semiconductor medium, or the like.

[0166] In the specification, claims, and accompanying drawings of this application, the terms “first”, “second”, and the like are intended to distinguish between similar objects but do not necessarily indicate a specific order or sequence. It should be understood that the terms used in such a way are interchangeable in proper circumstances, which is merely a discrimination manner that is used when objects having a same attribute are described in embodiments of this application. In addition, the terms “include”, “contain” and any other variants thereof mean to cover the non-exclusive inclusion, so that a process, method, system, product, or device that includes a series of

units is not necessarily limited to those units, but may include other units not expressly listed or inherent to the process, method, product, or device.

[0167] The terms used in embodiments of this application are merely for the purpose of describing specific embodiments, but are not intended to limit this application. “One” and “the” of singular forms used in embodiments of this application are also intended to include plural forms, unless the context clearly indicates another meaning. It should be further understood that, in the descriptions of this application, unless otherwise specified, “/” indicates an “or” relationship between associated objects. For example, A/B may represent A or B. “And/or” in this application describes only an association relationship between associated objects and represents that three relationships may exist. For example, A and/or B may represent the following three cases: Only A exists, both A and B exist, and only B exists. A and B may be singular or plural.

[0168] In conclusion, the foregoing embodiments are merely used to describe the technical solutions of this application, but not to limit this application. Although this application is described in detail with reference to the foregoing embodiments, persons of ordinary skill in the art should understand that they may still make modifications to the technical solutions recorded in the foregoing embodiments or make equivalent replacements to some technical features thereof, without departing from the scope of the technical solutions of embodiments of this application.

Claims

1. A beamforming apparatus, comprising a first laser signal source, M modulators, a first optical splitting device, N wavelength selection units, and N*M phase shifters, wherein the first laser signal source is configured to transmit M laser signals with different wavelengths; the M modulators are in a one-to-one correspondence with the M laser signals, and are respectively configured to modulate radio frequency signals on the corresponding laser signals to obtain M first modulated optical signals, wherein radio frequency signals modulated on any two of the M laser signals are the same or different; the first optical splitting device is coupled to the M modulators, and is configured to combine and split the M first modulated optical signals into N optical paths to obtain N second modulated optical signals corresponding to the N optical paths; the N wavelength selection units are coupled to the first optical splitting device; the N wavelength selection units are in a one-to-one correspondence with the N second modulated optical signals; a first wavelength selection unit is configured to: receive the second modulated optical signal sent by the first optical splitting device, and send signals with M wavelengths in the second modulated optical signal to M phase shifters in a one-to-one correspondence; and the first wavelength selection unit is any one of the N wavelength selection units; and the M phase shifters are respectively configured to: adjust phases of the received optical signals, and send the phase-adjusted optical signals to the first wavelength selection unit; and the first wavelength selection unit is further configured to fuse the received optical signals from the corresponding M phase shifters to obtain a third modulated optical signal.
2. The beamforming apparatus according to claim 1, wherein the wavelength selection unit is a reconfigurable optical add-drop multiplexer (ROADM).
3. The beamforming apparatus according to claim 1, wherein the beamforming apparatus further comprises: a control unit, configured to control phase shift spacings of the N*M phase shifters.
4. The beamforming apparatus according to claim 1, wherein the first laser signal source comprises M first laser devices, and the M first laser devices are coupled to the M modulators in a one-to-one correspondence; and wavelengths of the M first laser devices are different.
5. The beamforming apparatus according to claim 1, wherein the first laser signal source comprises an M-wavelength laser device and a wavelength division multiplexer, the M-wavelength laser device is coupled to the wavelength division multiplexer, and the wavelength division multiplexer is coupled to the M modulators.

6. The beamforming apparatus according to claim 1, wherein the beamforming apparatus further comprises N detectors and N radio frequency antenna units, the N detectors are coupled to the N wavelength selection units in a one-to-one correspondence, and the N detectors are coupled to the N radio frequency antenna units in a one-to-one correspondence; and the N detectors are configured to: receive third modulated optical signals sent by the corresponding wavelength selection units, perform optical-to-electrical conversion on the received third modulated optical signals to obtain electrical signals, and send the electrical signals through the radio frequency antenna units corresponding to the first detectors.

7. The beamforming apparatus according to claim 6, wherein the beamforming apparatus further comprises a second laser signal source and a second optical splitting device, wherein the second laser signal source is configured to transmit M carrier optical signals that are in a one-to-one correspondence with the M laser signals; the second optical splitting device is configured to couple and distribute the M carrier optical signals to the N detectors; and each of the N detectors is specifically configured to: perform coherent detection on the received third modulated optical signal based on received carrier optical signals, and perform optical-to-electrical conversion on the coherently detected third modulated optical signal to obtain an electrical signal.

8. The beamforming apparatus according to claim 6, wherein the beamforming apparatus further comprises a third optical splitting device, wherein one end of the third optical splitting device is coupled to the first laser signal source, the other end of the third optical splitting device is coupled to the N detectors, and the third optical splitting device is configured to couple and distribute the M laser signals to the N detectors; and a first detector in the N detectors is specifically configured to perform coherent detection on a received fourth modulated optical signal based on received laser signals, and perform optical-to-electrical conversion on a coherently detected fourth modulated optical signal to obtain an electrical signal.

9. A beamforming apparatus, comprising a first laser signal source, M modulators, M first beam splitters, N first wavelength division multiplexers, and $N \times M$ phase shifters, wherein the first laser signal source is configured to transmit M laser signals, wherein wavelengths of the M laser signals are different; the M modulators are in a one-to-one correspondence with the M laser signals, and are respectively configured to modulate radio frequency signals on the corresponding laser signals to obtain M first modulated optical signals, wherein radio frequency signals modulated on any two of the M laser signals are the same or different; the M first beam splitters are coupled to the M modulators in a one-to-one correspondence; and each of the M first beam splitters is coupled to input ends of N phase shifters, and is configured to: receive a first modulated optical signal sent by a corresponding modulator, distribute the received first modulated optical signal into N second modulated optical signals, and distribute the N second modulated optical signals to the N phase shifters in a one-to-one correspondence; the N phase shifters are respectively configured to adjust phases of the received second modulated optical signals to obtain third modulated optical signals; and an input end of each of the N first wavelength division multiplexers is coupled to output ends of M phase shifters, different first beam splitters are respectively coupled to input ends of M phase shifters coupled to a same first wavelength division multiplexer, and each first wavelength division multiplexer is configured to couple third modulated optical signals from the M phase shifters into a fourth modulated optical signal.

10. The beamforming apparatus according to claim 9, wherein the beamforming apparatus further comprises: a control unit, configured to control phase shift spacings of the $N \times M$ phase shifters.

11. The beamforming apparatus according to claim 9, wherein the first laser signal source comprises M first laser devices, and the M first laser devices are coupled to the M modulators in a one-to-one correspondence; and wavelengths of the M first laser devices are different.

12. The beamforming apparatus according to claim 9, wherein the first laser signal source comprises a M-wavelength laser device and a second wavelength division multiplexer, the M-wavelength laser device is coupled to the second wavelength division multiplexer, and the second

wavelength division multiplexer is coupled to the M modulators.

13. The beamforming apparatus according to claim 9, wherein the beamforming apparatus further comprises N detectors and N radio frequency antenna units; first ends of the N detectors are coupled to second ends of the N first wavelength division multiplexers in a one-to-one correspondence, and second ends of the N detectors are coupled to the N radio frequency antenna units in a one-to-one correspondence; and each of the N detectors is configured to: receive a fourth modulated optical signal from a corresponding first wavelength division multiplexer, perform optical-to-electrical conversion on the fourth modulated optical signal to obtain an electrical signal, and send the electrical signal through a corresponding radio frequency antenna unit.

14. The beamforming apparatus according to claim 13, wherein the beamforming apparatus further comprises a second laser signal source and a second optical splitting device, wherein the second laser signal source is configured to transmit M carrier optical signals that are in a one-to-one correspondence with the M laser signals; the second optical splitting device is configured to couple and distribute the M carrier optical signals to the N detectors; and each of the N detectors is specifically configured to: perform coherent detection on the received fourth modulated optical signal based on received carrier optical signals, and perform optical-to-electrical conversion on the coherently detected fourth modulated optical signal to obtain an electrical signal.

15. The beamforming apparatus according to claim 13, wherein the beamforming apparatus further comprises a third optical splitting device, wherein one end of the third optical splitting device is coupled to the first laser signal source, the other end of the third optical splitting device is coupled to the N detectors, and the third optical splitting device is configured to couple and distribute the M laser signals to the N detectors; and a first detector in the N detectors is specifically configured to: perform coherent detection on a received fourth modulated optical signal based on received laser signals, and perform optical-to-electrical conversion on a coherently detected fourth modulated optical signal to obtain an electrical signal.

16. A beamforming method, wherein the method comprises: transmitting M laser signals with different wavelengths; respectively modulating radio frequency signals on the M laser signals to obtain M first modulated optical signals, wherein radio frequency signals modulated on any two of the M laser signals are the same or different; combining and distributing the M first modulated optical signals into N optical paths, to obtain N second modulated optical signals corresponding to the N optical paths; and on each optical path, performing phase adjustment on optical signals with M wavelengths comprised in the second modulated optical signal, and coupling and performing optical-to-electrical conversion on the M phase-shifted optical signals to obtain an electrical signal.

17. The method according to claim 16, wherein the method further comprises: transmitting M carrier optical signals that are in a one-to-one correspondence with the M laser signals; and combining and splitting the M carrier optical signals into N paths of optical signals, wherein each of the N paths of optical signals comprises M carrier optical signals; and the coupling and performing optical-to-electrical conversion on the M phase-shifted optical signals to obtain an electrical signal comprises: on each optical path, respectively performing coherent detection on the M phase-shifted optical signals based on M received carrier optical signals, and performing optical-to-electrical conversion on the coherently detected optical signals to obtain the electrical signal.

18. The method according to claim 16, wherein the method further comprises: coupling and splitting the M laser signals with the different wavelengths into N paths of optical signals, wherein the N paths of optical signals comprises M laser signals with the different wavelengths; and the coupling and performing optical-to-electrical conversion on the M phase-shifted optical signals to obtain an electrical signal comprises: respectively performing coherent detection on the M phase-shifted optical signals based on M received laser signals, and performing optical-to-electrical conversion on the coherently detected optical signals to obtain the electrical signal.
